Lesson 04 Demo 04

Demonstrating Programmatic Navigation in a React Application

Objective: To demonstrate programmatic navigation in a React application using Vite to improve user redirection and navigation experience

Tools required: Visual Studio Code and Node.js

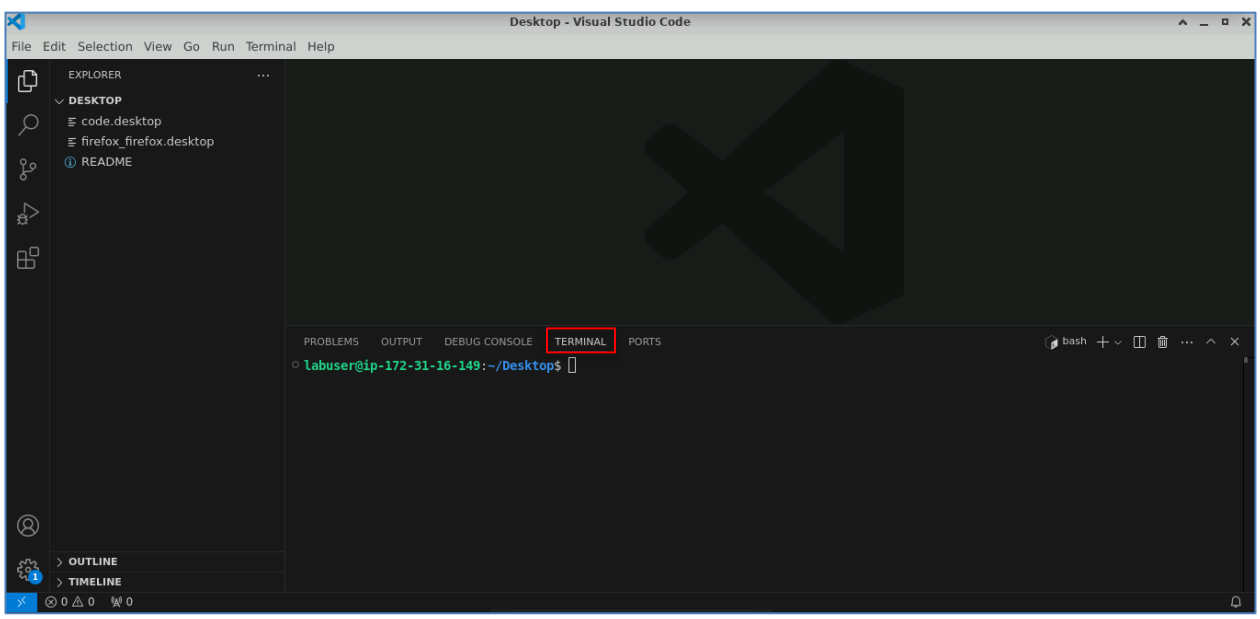
Prerequisites: None

Steps to be followed:

1. Set up a new React project using Vite
2. Create React components
3. Import and use the components
4. Run and verify the application

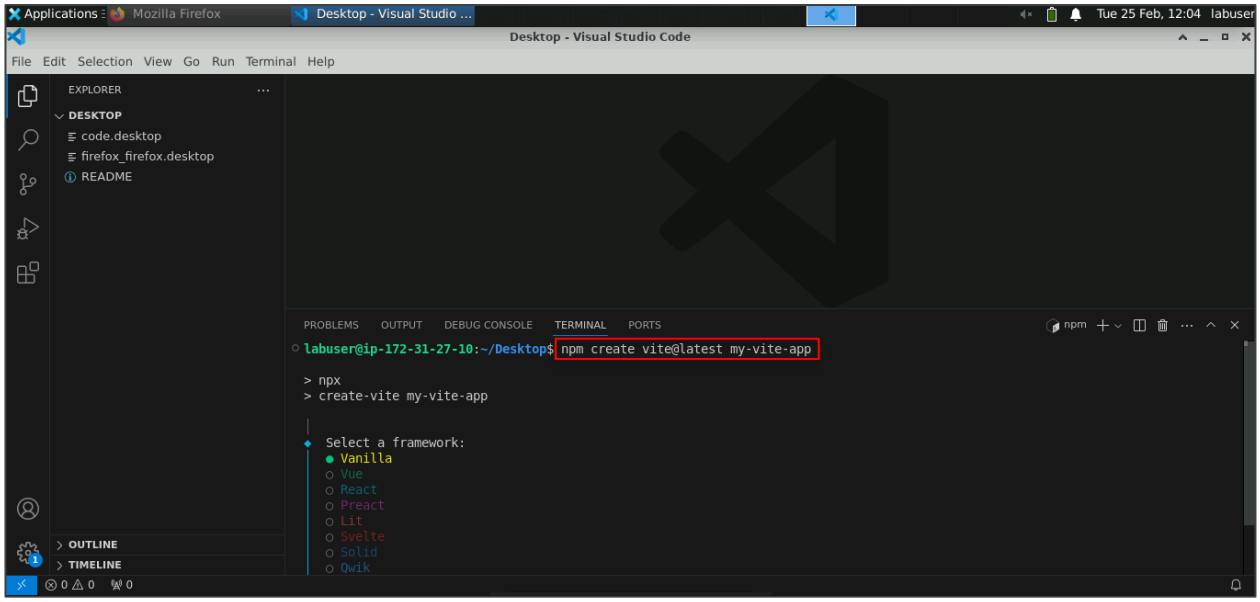
Step 1: Set up a new React project using Vite

1.1 Navigate to VS Code and open the terminal

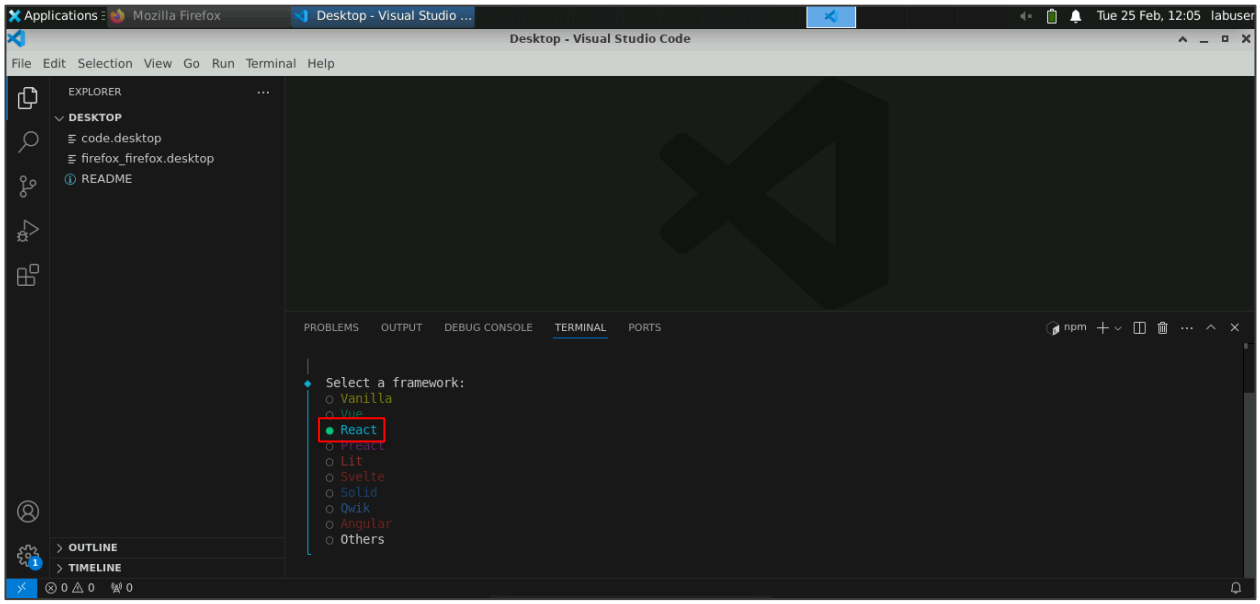


1.2 Run the following command to create a new React project using Vite:

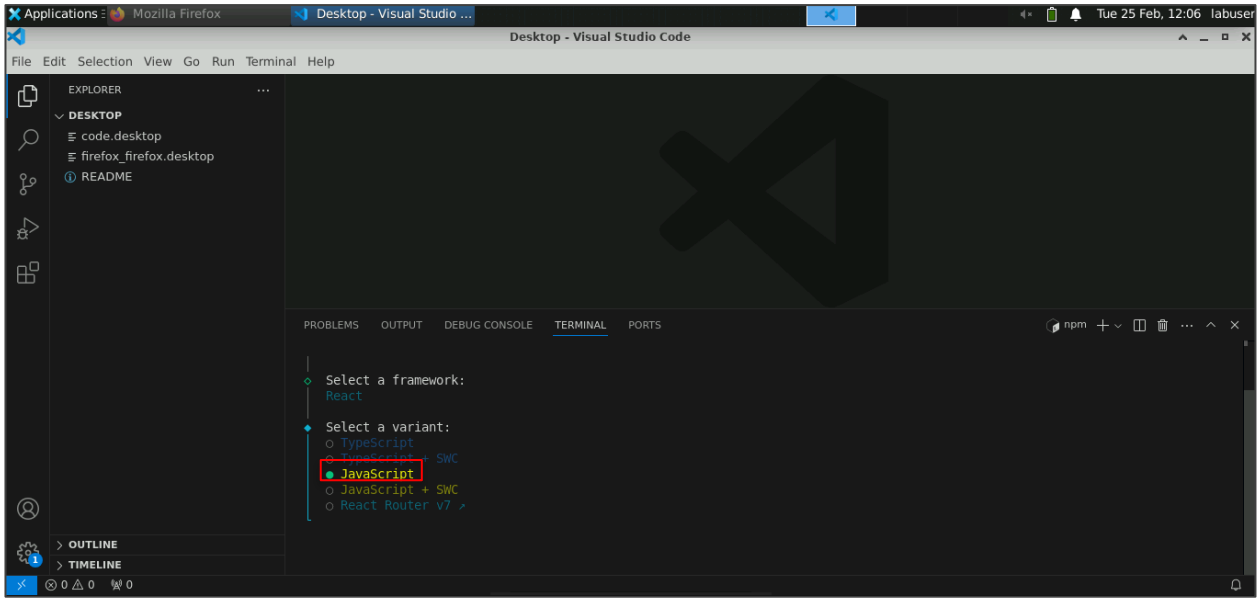
npm create vite@latest my-vite-app



1.3 Select **React** and press enter

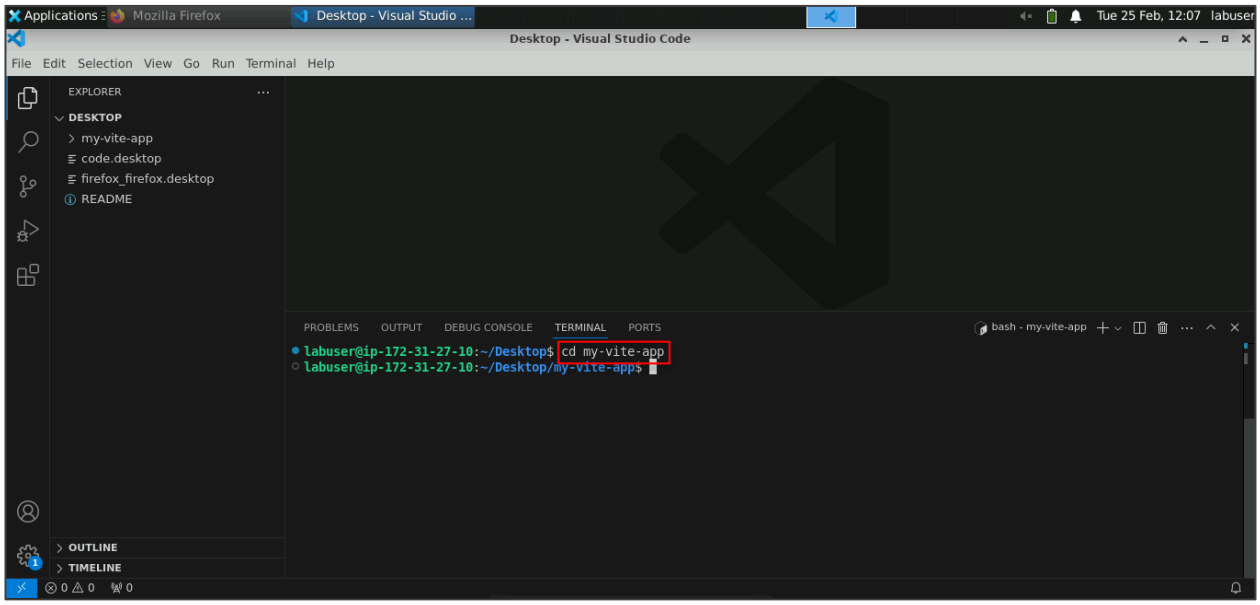


1.4 Select **JavaScript** and press enter



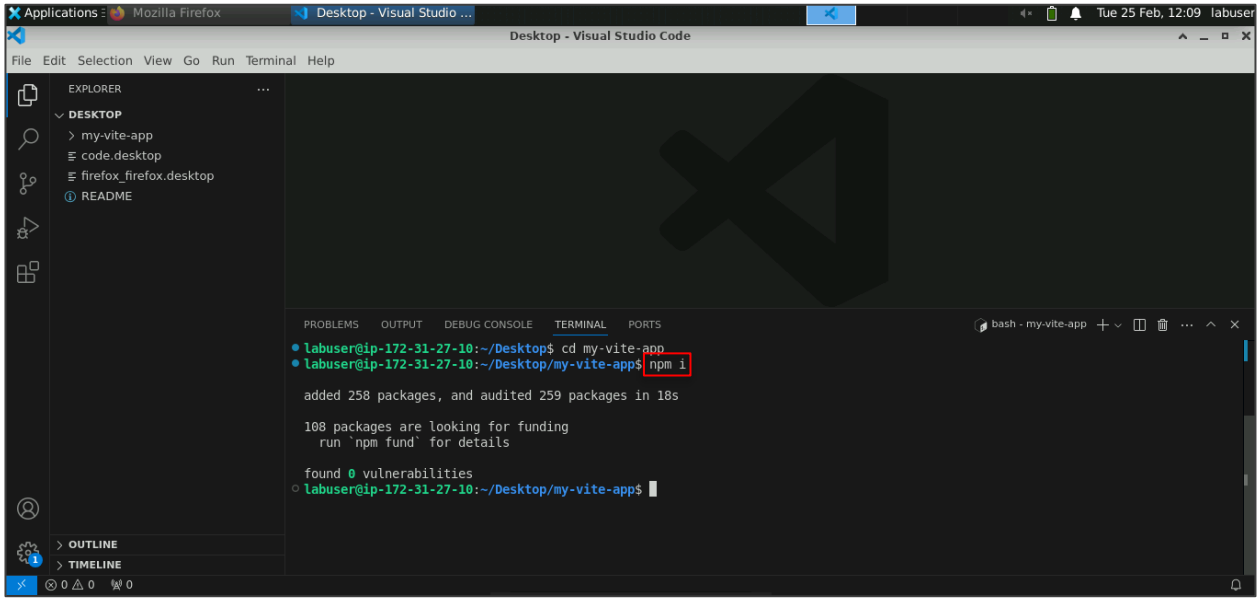
1.5 Run the following command to navigate to the project folder:

cd my-vite-app



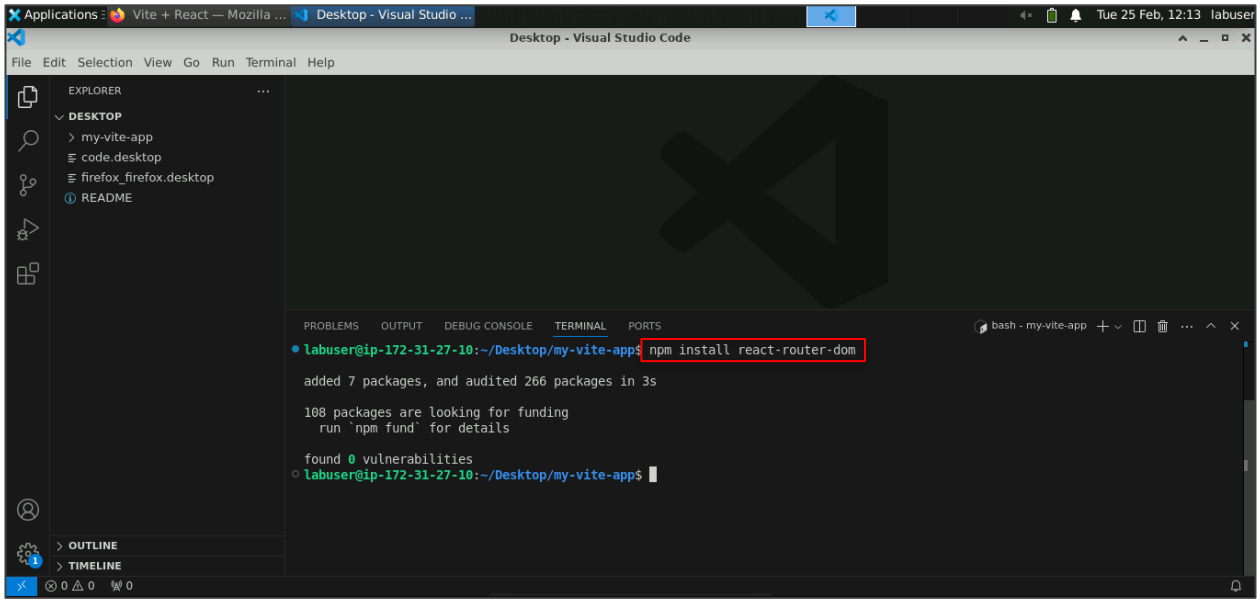
1.6 Run the following command to install dependencies:

npm i



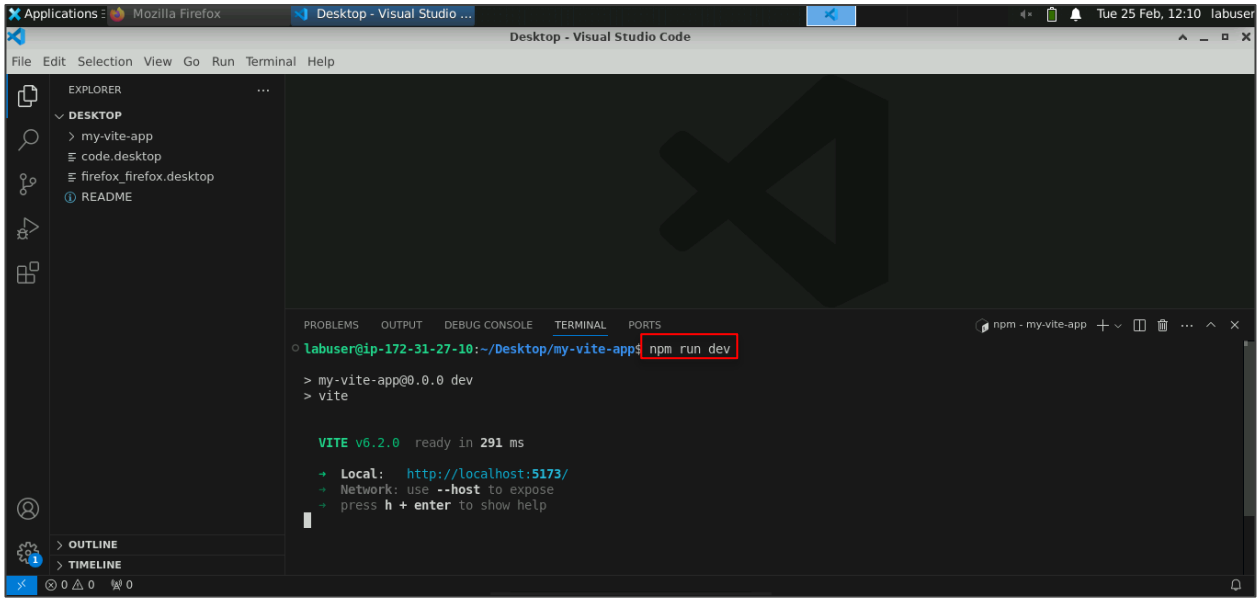
1.7 Run the following command to install the React router:

npm install react-router-dom



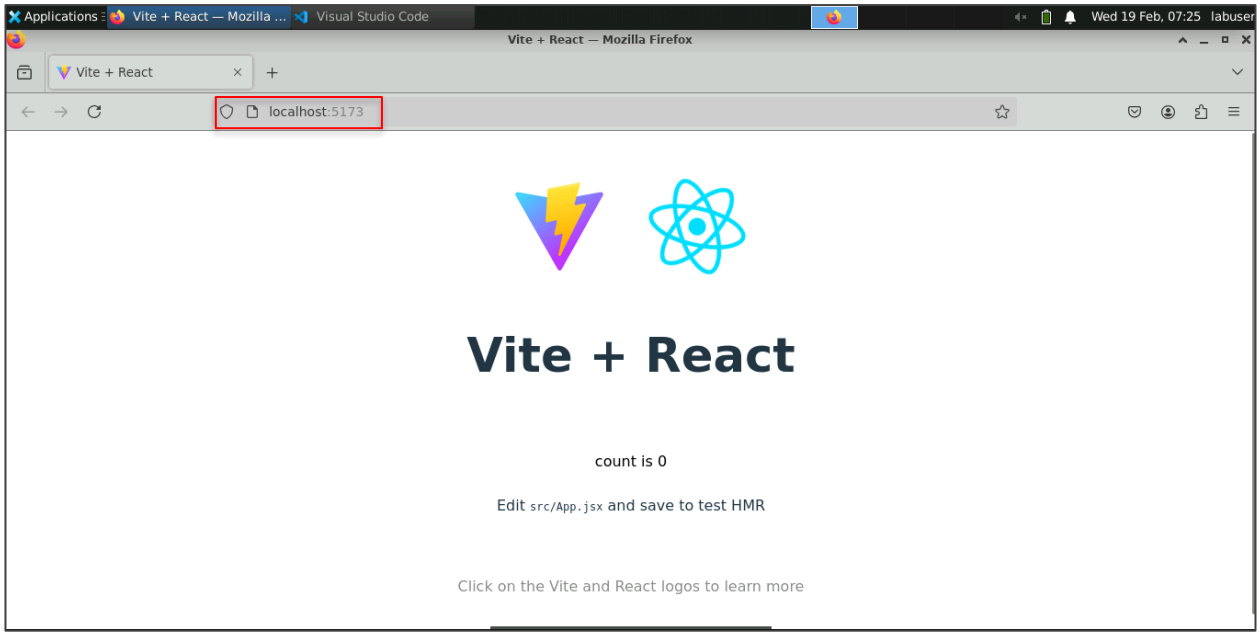
1.8 Run the following command to start the development server:

npm run dev



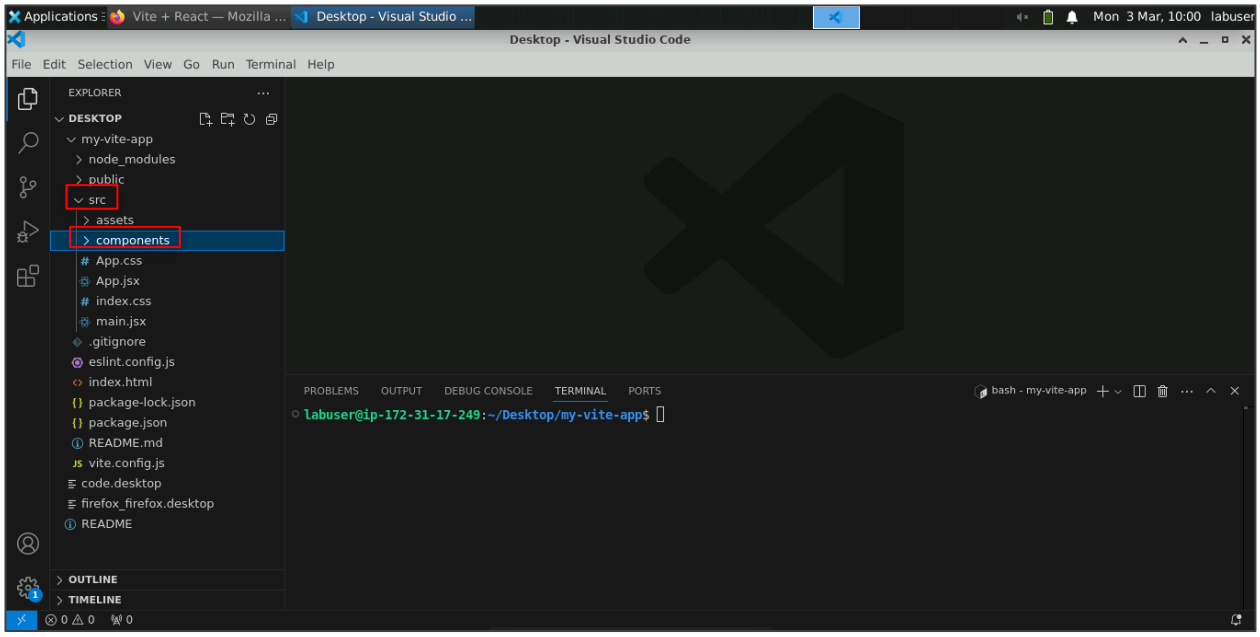
1.9 Navigate to the web browser and paste the link below:

http://localhost:5173

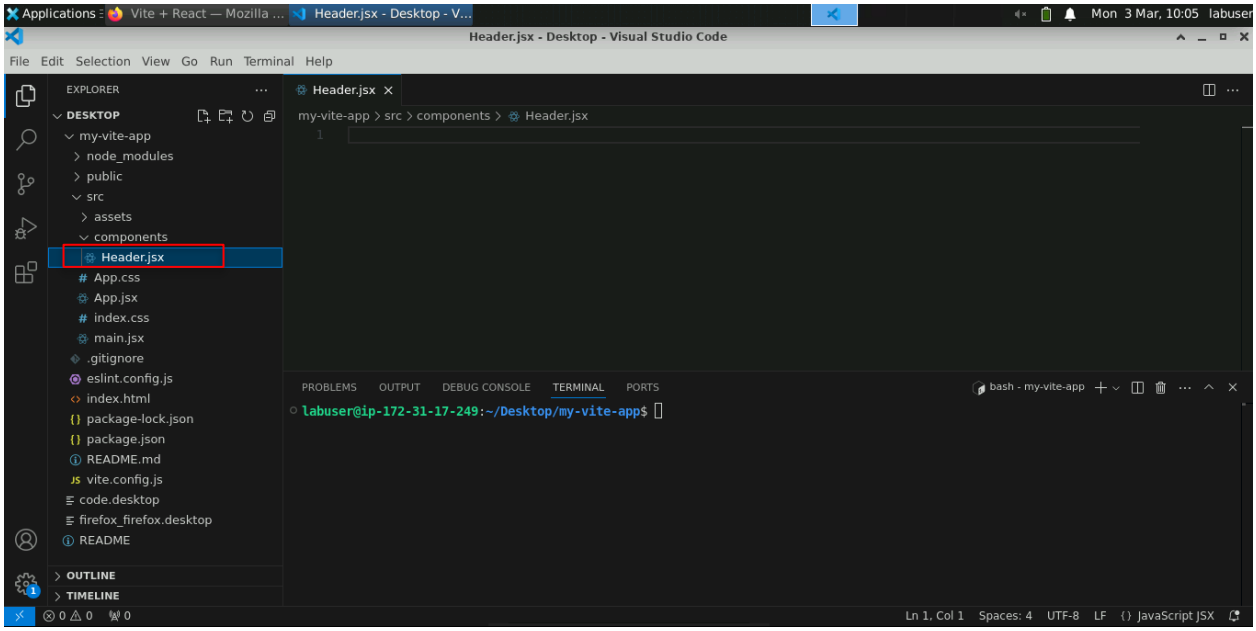


Step 2: Create React components

2.1 Inside the **src** folder, create a new folder named **components**

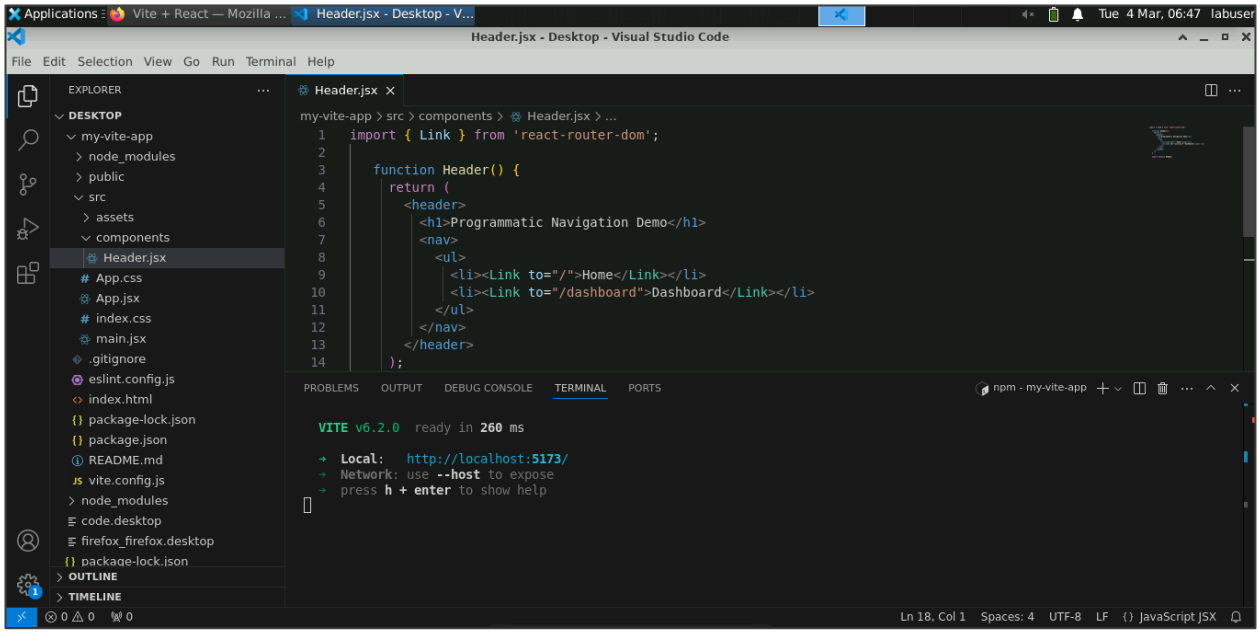


2.2 Inside the **components** folder, create a new file named **Header.jsx**

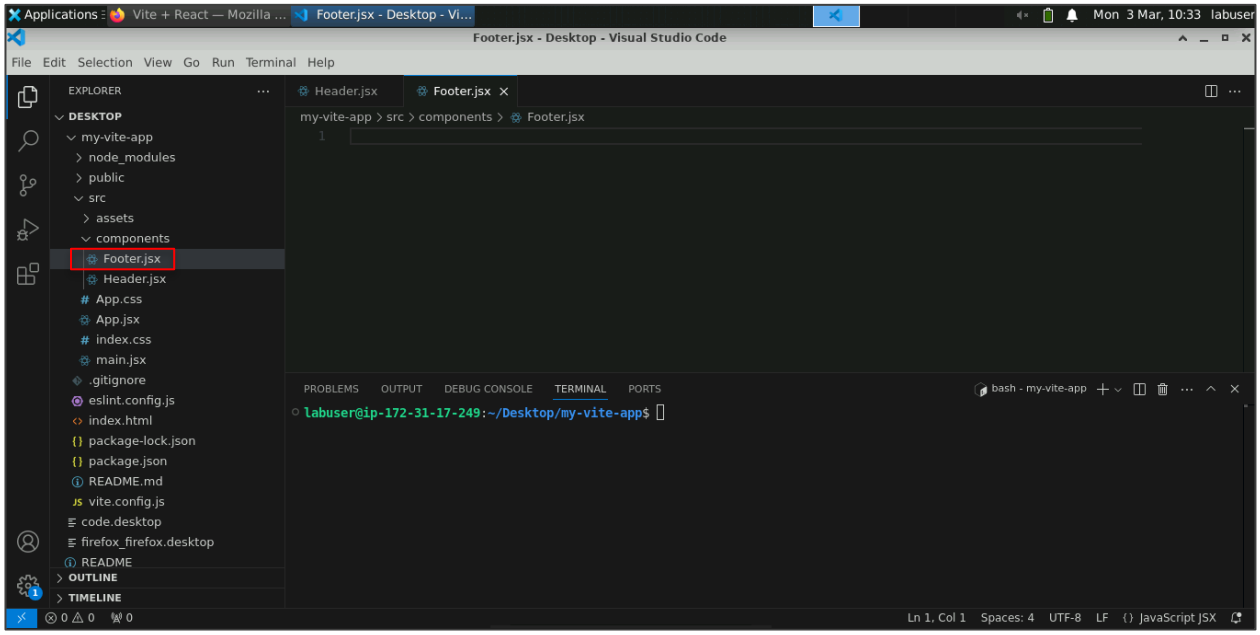


2.3 Add the following code to **Header.jsx:**

```
import { Link } from 'react-router-dom';

  function Header() {
   return (
    <header>
     <h1>Programmatic Navigation Demo</h1>
     <nav>
      <ul>
       <li><Link to="/">Home</Link></li>
       <li><Link to="/dashboard">Dashboard</Link></li>
      </ul>
     </nav>
    </header>
   );
  }

  export default Header;
```

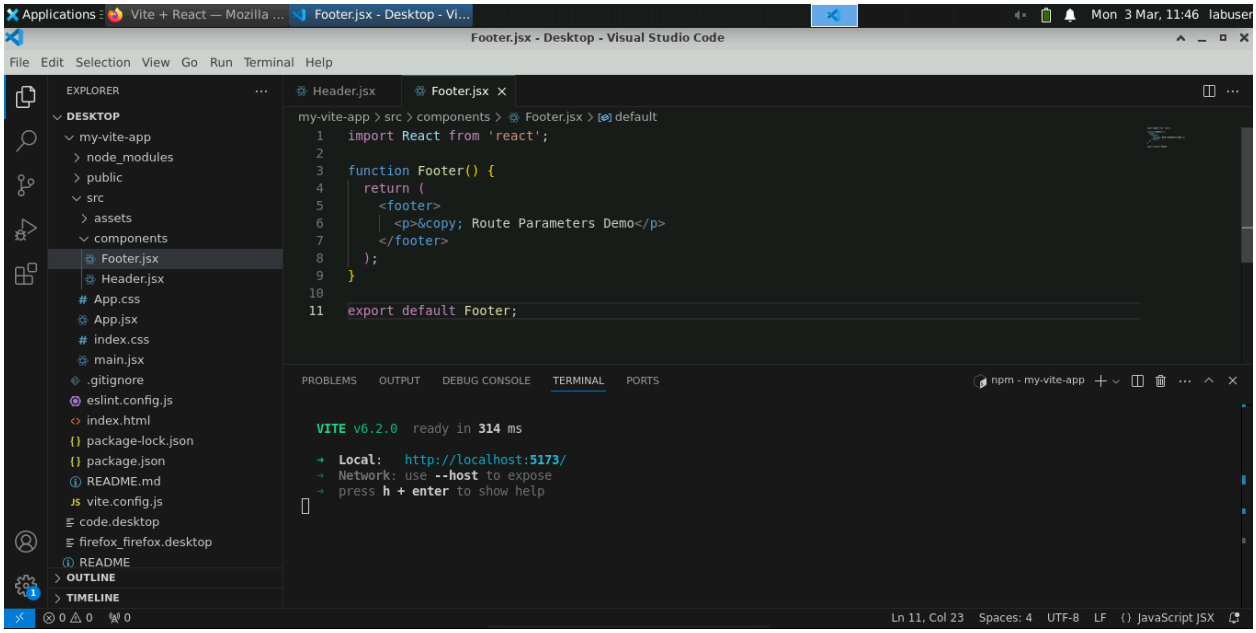


2.4 Inside the **components** folder, create a new file named **Footer.jsx**

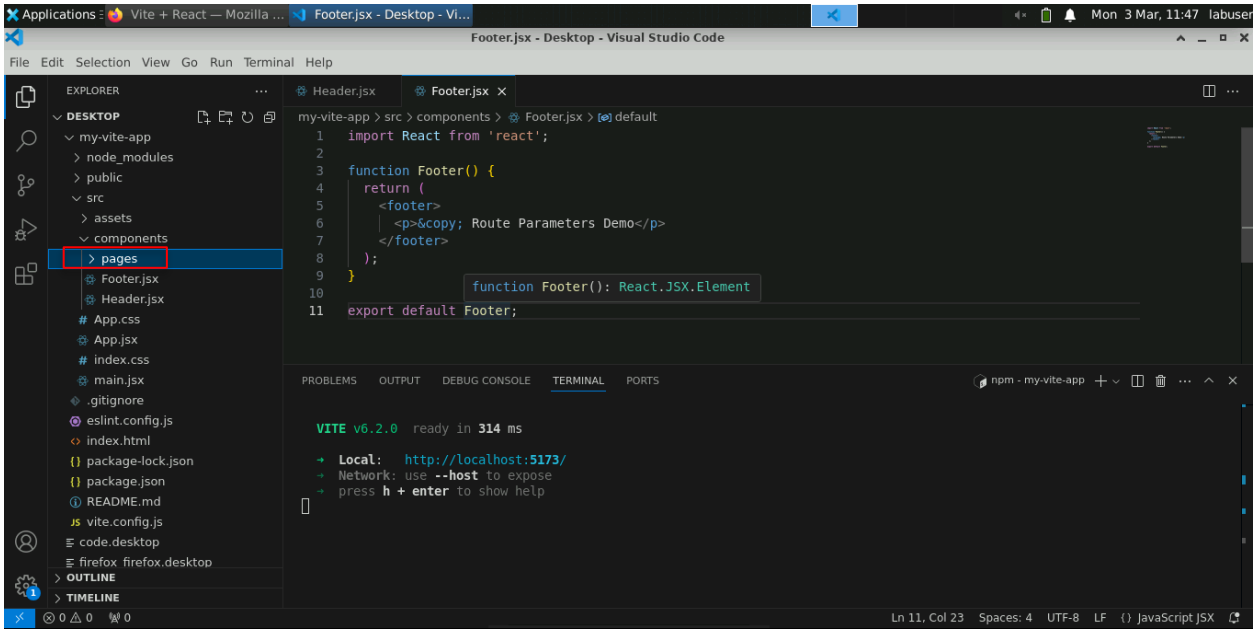


2.5 Add the following code to **Footer.jsx:**

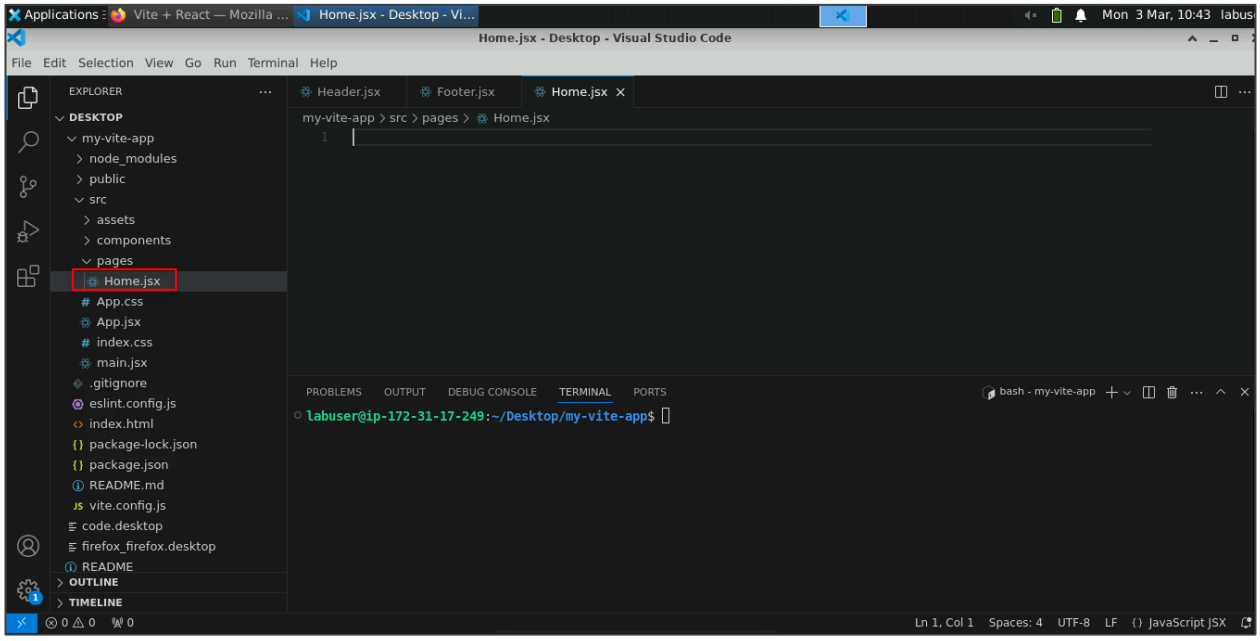
```
import React from 'react';

  function Footer() {
    return (
      <footer>
        <p>© Programmatic Navigation</p>
      </footer>
    );
  }
export default Footer;
```



2.6 Inside the **src** folder, create a new folder named **pages**

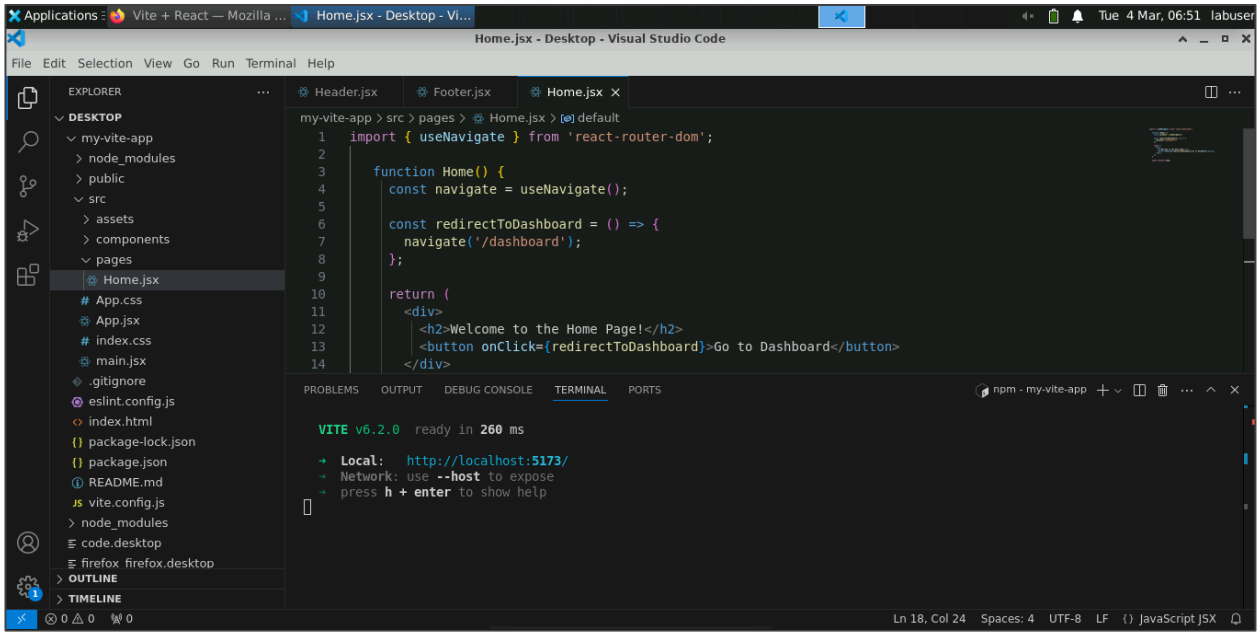


2.7 Inside the **pages** folder, create a new file named **Home.jsx**

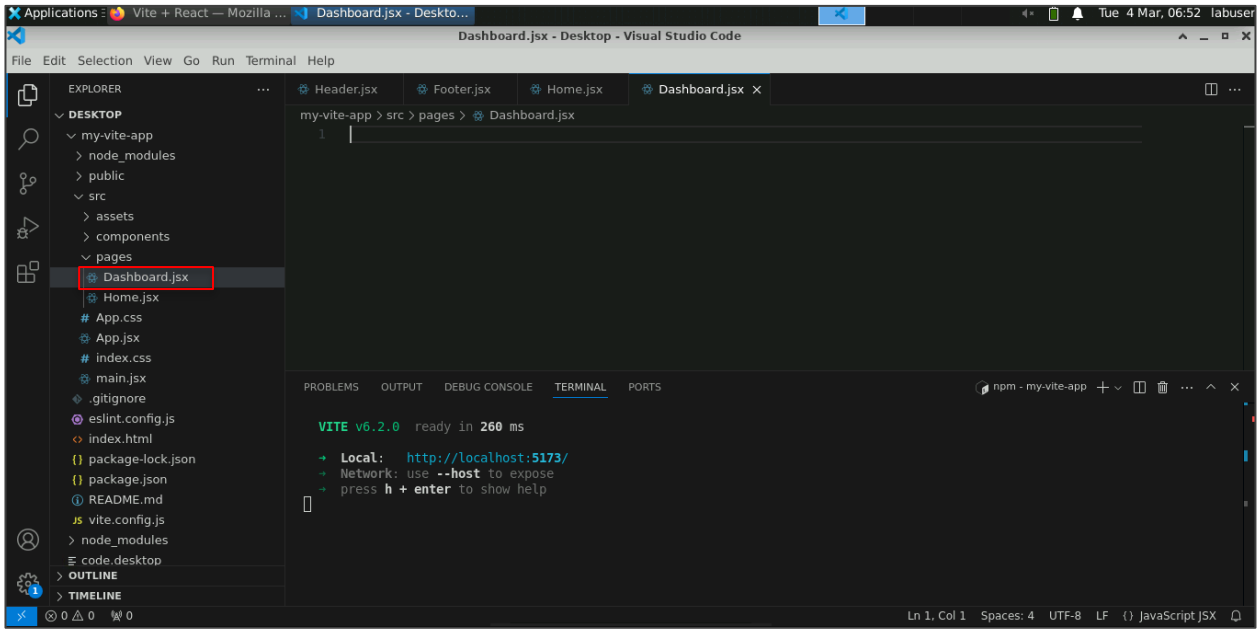


2.8 Add the following code to **Home.jsx:**

```
import { useNavigate } from 'react-router-dom';

  function Home() {
    const navigate = useNavigate();

    const redirectToDashboard = () => {
      navigate('/dashboard');
    };

    return (
      <div>
        <h2>Welcome to the Home Page!</h2>
        <button onClick={redirectToDashboard}>Go to Dashboard</button>
      </div>
    );
  }

  export default Home;
```

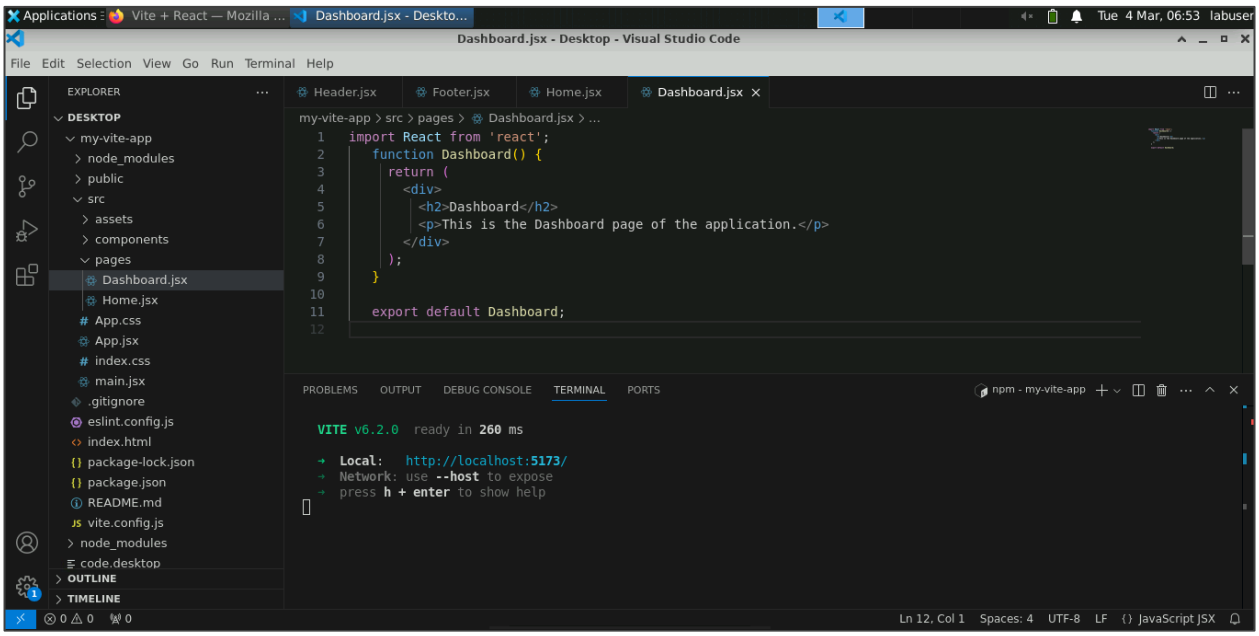


2.9 Inside the **pages** folder, create a new file named **Dashboard.jsx**



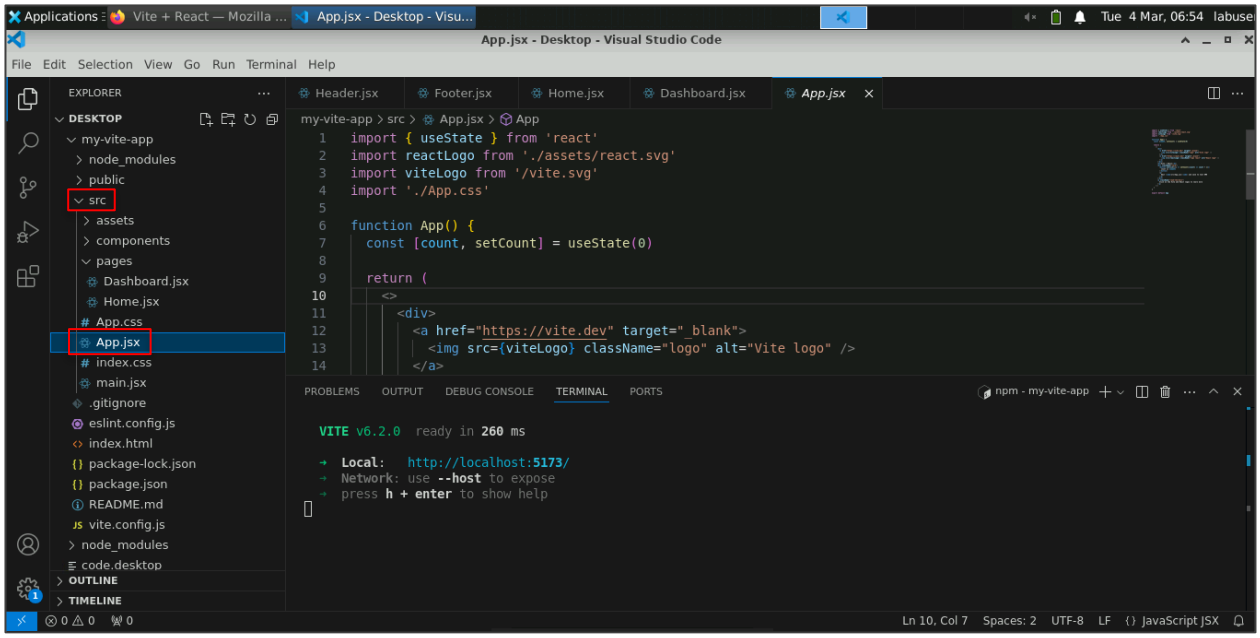
2.10 Add the following code to **Dashboard.jsx:**

```
import React from 'react';
  function Dashboard() {
    return (
      <div>
        <h2>Dashboard</h2>
        <p>This is the Dashboard page of the application.</p>
      </div>
    );
  }

  export default Dashboard;
```



Step 3: Import and use the components

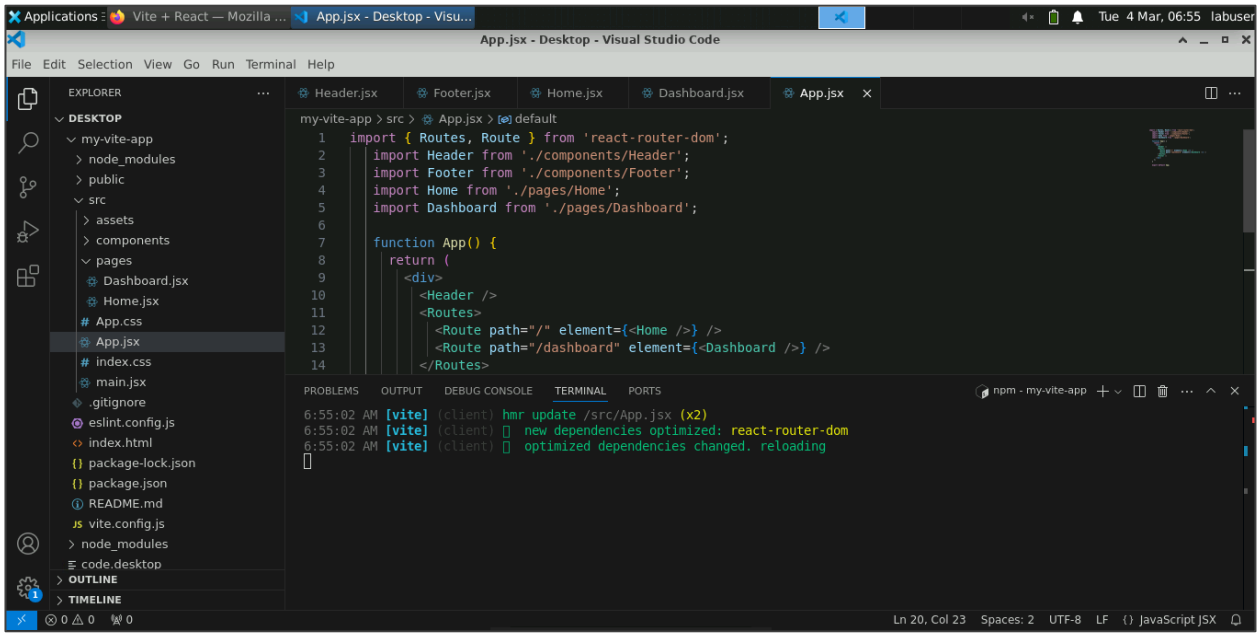
3.1 Open the **App.jsx** file inside the **src** folder



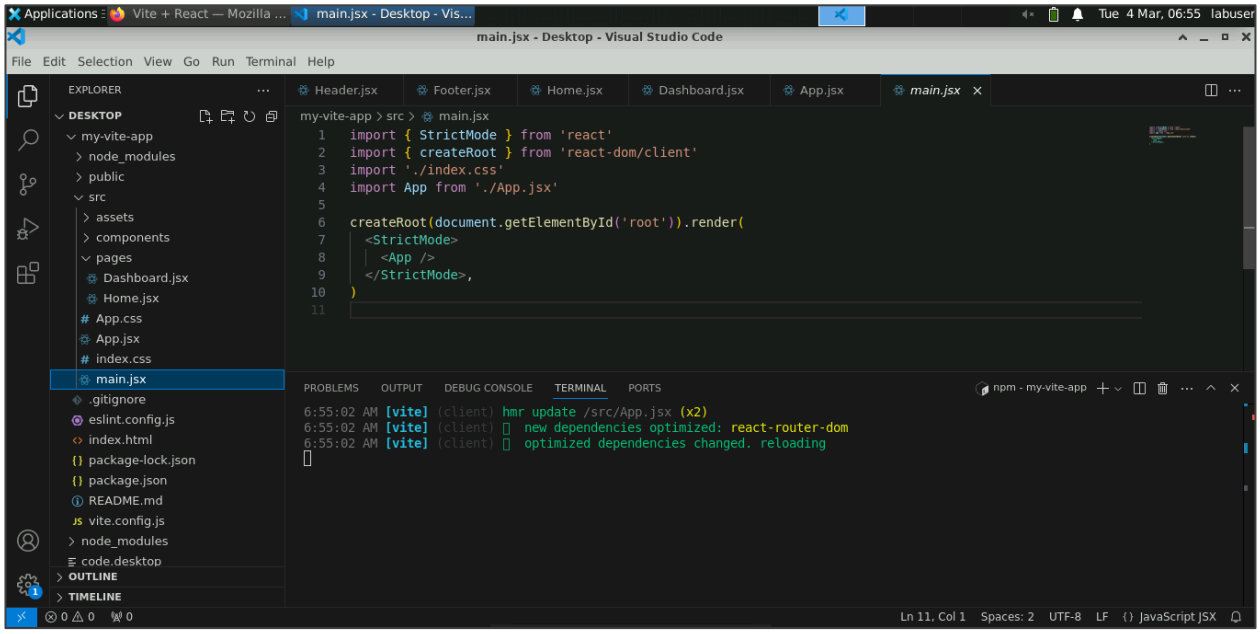
3.2 Paste the code below to include the new components in **App.jsx**:

```
import { Routes, Route } from 'react-router-dom';
  import Header from './components/Header';
  import Footer from './components/Footer';
  import Home from './pages/Home';
  import Dashboard from './pages/Dashboard';

  function App() {
    return (
      <div>
        <Header />
        <Routes>
          <Route path="/" element={<Home />} />
          <Route path="/dashboard" element={<Dashboard />} />
        </Routes>
        <Footer />
      </div>
    );
```

```
}

export default App;
```

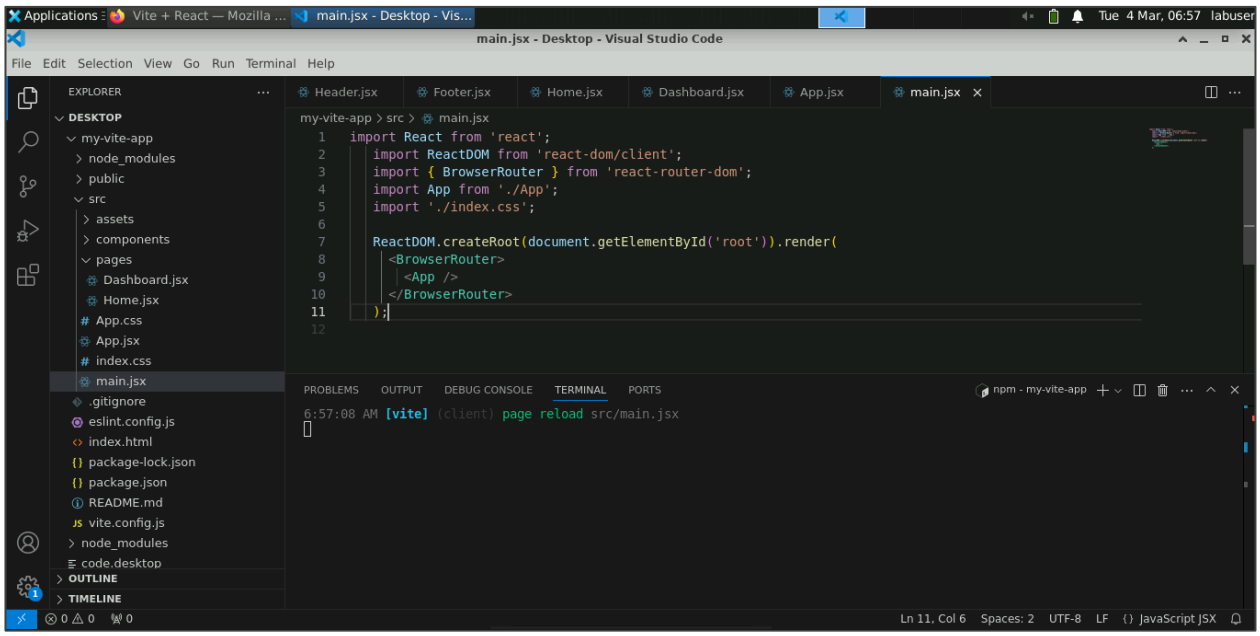


3.3 Open the **main.jsx** file inside the **src** folder



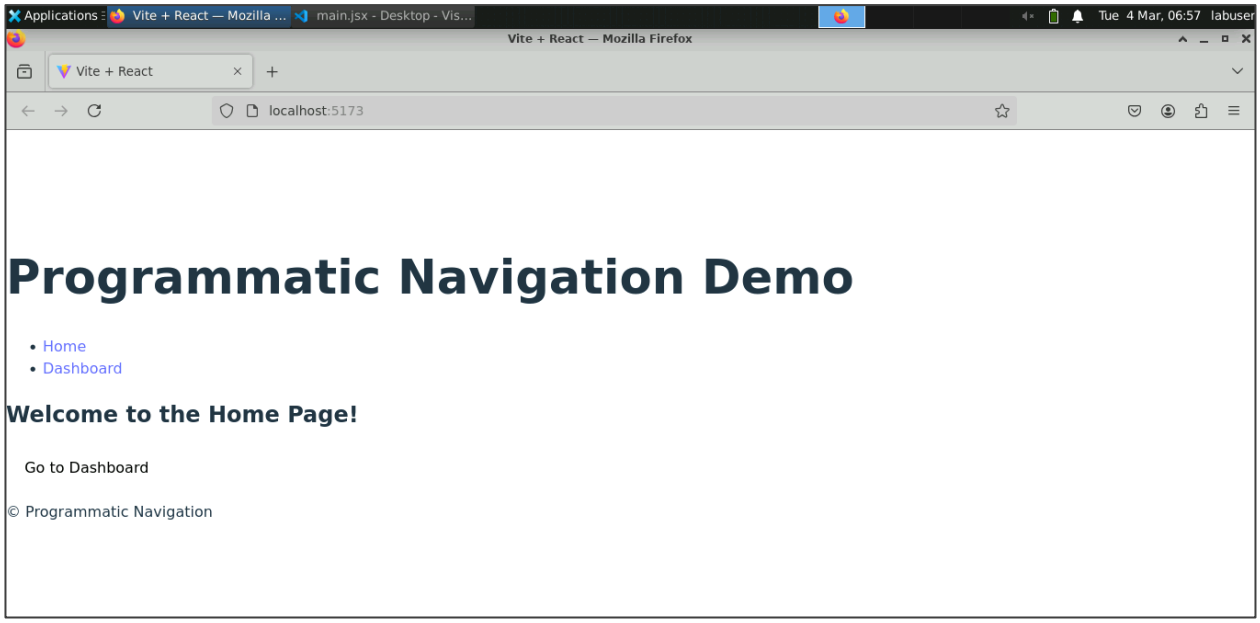
3.4 Paste the code below to include the new components in **main.jsx**:

```
import React from 'react';
  import ReactDOM from 'react-dom/client';
  import { BrowserRouter } from 'react-router-dom';
  import App from './App';
  import './index.css';

  ReactDOM.createRoot(document.getElementById('root')).render(
    <BrowserRouter>
      <App />
    </BrowserRouter>
  );
```

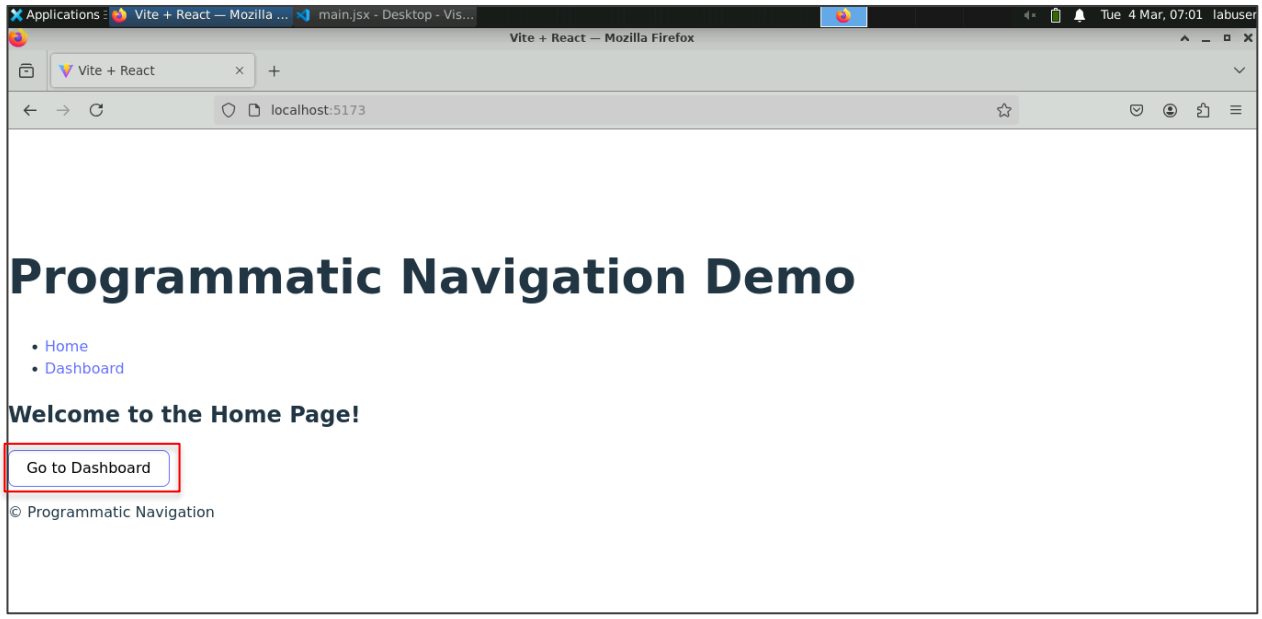


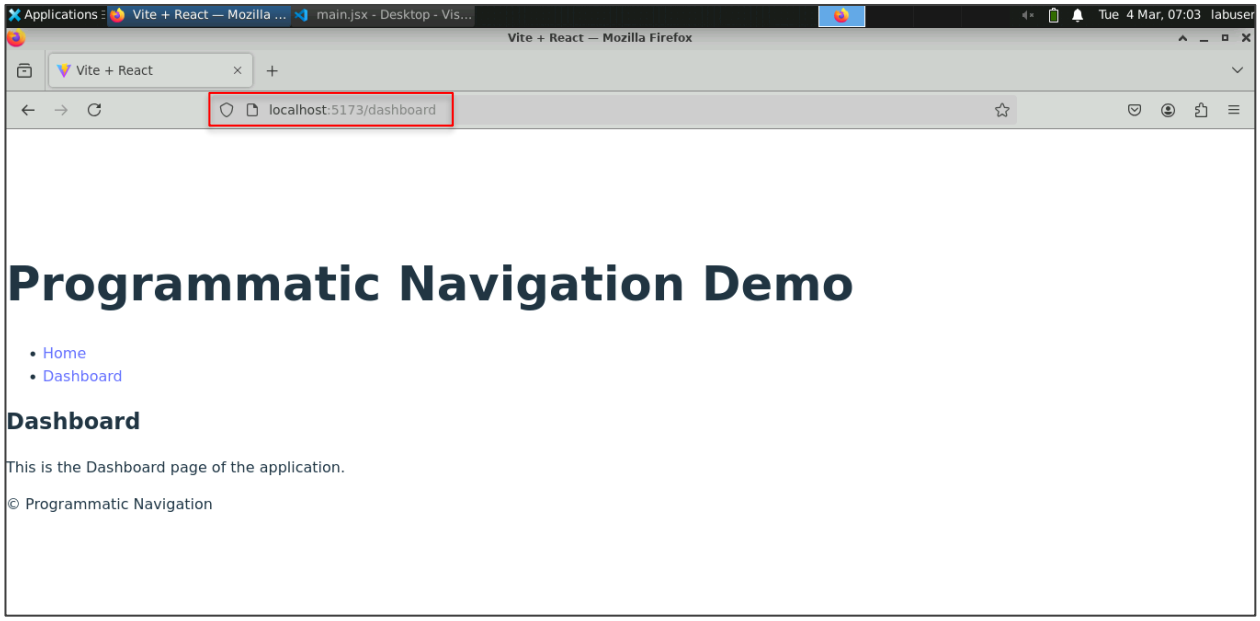
Step 4: Run and verify the application

4.1 Navigate to the browser and refresh the webpage



4.2 Click **Go to Dashboard** to navigate to **/dashboard** route





By following the above steps, you have successfully implemented programmatic navigation in a React app using Vite by setting up routing, creating components, and using useNavigate for redirection, providing a seamless navigation experience.